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Ruei-Che Chang

Education

2022 – Present **University of Michigan**, *Ann Arbor, Michigan*
Ph.D. Candidate in Computer Science & Engineering.
Human-AI Lab, advised by Anhong Guo

2020 – 2022 **National Taiwan University**, *Taipei, Taiwan*
Visiting Student.
Interactive Graphics Lab, advised by Bing-Yu Chen

2019 – 2021 **Dartmouth College**, *Hanover, New Hampshire*
M.S. in Computer Science.

2014 – 2018 **National Cheng Kung University**, *Tainan, Taiwan*
B.S. in Electrical Engineering.

Professional Experiences

2026 **Apple Inc.**, *Seattle, Washington*
May – Aug **AI/ML Intern**. Host: Cole Gleason
- Explored novel agentic interactions for visual question answering.

2025 **Adobe Research**, *Seattle, Washington*
May – Aug **Research Scientist Intern**. Host: Bryan Wang and Dingzeyu Li
- Explored communication between human and computer use agent [C.21].

2024 **Meta Reality Labs Research**, *Toronto, Ontario, Canada*
May – Aug **Research Scientist Intern**. Host: Hemant Surale
- Explored visual-audio modality transitions for mobile tasks on the go [C.18].

Honors, Awards and Fellowships

2026 **Taiwan Government Scholarship**
One million New Taiwanese dollars (USD $32000) for two years

Rackham Predoctoral Fellowship
Full tuition and stipend coverage for one academic year

2025 **Apple Scholars in AI/ML PhD Fellowship (AI for Accessibility)**
Full tuition, stipend coverage, and travel support for two years (2025-2027)

UIST 2025 Doctoral Symposium, Busan, Korea

ASSETS 2025 Doctoral Consortium, Denver, Colorado

2024 **Best Paper Award at ACM UIST 2024 for WorldScribe [C.15]**
Top 4 out of 608 submissions

Google Academic Research Award for WorldScribe [C.15]
Led a proposal with Prof. Anhong Guo (Faculty-PI) resulting in $75000 resesarch funding

Weinberg Cognitive Science Fellowship
Full tuition and stipend coverage for one semester

Finalist, CSE Honors Competition, University of Michigan
One of the five finalists recognized as "top research done by PhD students" at CSE.

2023 **Rackham International Students Chia-Lun Lo Fellowship**
$13,770 for stipend in Summer 2024

2022 **University of Michigan CSE Departmental Fellowship**

Full tuition and stipend coverage for first-year PhD Study

2020 **Best Paper Honorable Mention at ACM CHI 2020 for Glissade [C.2]**
Top 5% out of 3216 submissions

2019–2021 **Dartmouth College Tuition Scholarship**
75% tuition coverage for master's study

2023–2026 **Special Recognition for Outstanding Reviews × 11**
CHI '23 '24 '25 '26, **UIST** '23 '24 '25, **DIS** '24

2022–2026 **Rackham Travel Grant Awards**
UIST'22, UIST'23, ASSETS'24, UIST'25, UIST'26

Peer-Reviewed Full Papers

2026 [C.22] **Ruei-Che Chang***, Xirui Jiang*, Rosiana Natalie, Hao Chen, Vlad Roznyatovskiy, Jianzhong Zhang, Kang G. Shin, Ke Sun, Anhong Guo. "StateScribe: Towards Accessible Change Awareness Across Real-World Revisits." *In The 39th Annual ACM Symposium on User Interface Software and Technology* (***UIST'26***). *Detroit, MI. 2026.* [*Acceptance Rate: 20.1%*]

[C.21] **Ruei-Che Chang**, Wenqian Xu, Dingzheyu Li, Bryan Wang, Anhong Guo. "Sidekick: Designing Communication for Effective Multitasking with Computer Use Agents." *In The 39th Annual ACM Symposium on User Interface Software and Technology* (***UIST'26***). *Detroit, MI. 2026.* [*Acceptance Rate: 20.1%*]

[C.20] **Ruei-Che Chang**, Rosiana Natalie, Wenqian Xu, Jovan Zheng Feng Yap, Tiange Luo, Venkatesh Potluri, Anhong Guo. "TouchScribe: Augmenting Hand-Object Interaction with Automated Live Visual Descriptions." *In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (***CHI'26***). *Barcelona, Spain. 2026.* [*Acceptance Rate: 25.3%*]

2025 [C.19] **Ruei-Che Chang**, Rosiana Natalie, Wenqian Xu, Jovan Zheng Feng Yap, Anhong Guo. "Probing the Gaps in ChatGPT Live Video Chat for Real-World Assistance for People who are Blind or Visually Impaired." *In Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility* (***ASSETS'25***). *Denver, Colorado. 2025.* [*Acceptance Rate: 29.7%*]

[C.18] **Ruei-Che Chang**, Tovi Grossman, Carine Rognon, Michael Glueck, Christopher Collins, Amy Karlson, Hemant Bhaskar Surale. "Viago: Exploring Visual-Audio Modality Transitions for Social Media Consumption On the Go." *In The 38th Annual ACM Symposium on User Interface Software and Technology* (***UIST'25***). *Busan, Korea. 2025.* [*Acceptance Rate: 22%*]

[C.17] Rosiana Natalie, Wenqian Xu, **Ruei-Che Chang**, Rada Mihalcea, Anhong Guo. "Not There Yet: Evaluating Vision Language Models in Simulating the Visual Perception of People with Low Vision." *Preprint*

[C.16] Yu-Ting Yen, Fang-Ying Liao, Chi-Lan Yang, **Ruei-Che Chang**, Fu-Yin Cherng, Bing-Yu Chen. "Strange Familiars: Exploring the Design of Avatars and Virtual Environments for Reconnecting Dormant Ties in Virtual Reality." *IEEE Transactions on Visualization and Computer Graphics* (***TVCG'25***). *2025.*

2024 [C.15] **Ruei-Che Chang**, Yuxuan Liu, Anhong Guo. "WorldScribe: Towards Context-Aware Live Visual Descriptions." *In The 37th Annual ACM Symposium on User Interface Software and Technology* (***UIST'24***). *Pittsburgh, PA, USA. 2024.* [*Acceptance Rate: 24%*] **Best Paper Award (Top 4 of 608 submissions).**

[C.14] **Ruei-Che Chang**, Chia-Sheng Hung, Bing-Yu Chen, Dhruv Jain, Anhong Guo. "SoundShift: Exploring Sound Manipulations for Accessible Mixed-Reality Awareness." *In Proceedings of the 2024 ACM Conference on Designing Interactive Systems* (***DIS'24***). *Copenhagen, Denmark. 2024.*

[C.13] **Ruei-Che Chang**, Yuxuan Liu, Lotus Zhang, Anhong Guo. "EditScribe: Non-Visual Image Editing with Natural Language Verification Loop." *In Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility* (***ASSETS'24***). *St. John's, Newfoundland, Canada. 2024.* [*Acceptance Rate: 30%*]

[C.12] Rosiana Natalie, **Ruei-Che Chang**, Smitha Sheshadri, Anhong Guo, Kotaro Hara. "Audio Description Customization." *In Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility* (***ASSETS'24***). *St. John's, Newfoundland, Canada. 2024.* [*Acceptance Rate: 30%*]

[C.11] Andi Xu, Minyu Cai, Dier Hou, **Ruei-Che Chang**, Anhong Guo. "ImageExplorer Deployment: Understanding Text-Based and Touch-Based Image Exploration in the Wild." *In Proceedings of the 21st Web for All Conference* (***W4A 2024***). *Sentosa, Singapore. 2024.*

[C.10] Hao-Ping Lee, Wei-Lun Kao, Hung-Jui Wang, **Ruei-Che Chang**, Yi-Hao Peng, Fu-Ying Cherng, Shang-Tse Chen. "AdvCAPTCHA: Creating Usable and Secure Audio CAPTCHA with Adversarial Machine Learning." *NDSS Symposium on Usable Security and Privacy* (***USEC'24***). *San Diego, California. 2024.*

2023 [C.9] **Ruei-Che Chang***, Seraphina Yong*, Fang-Ying Liao, Chih-An Tsao, Bing-Yu Chen. "Understanding (Non-)Visual Needs of the Design of Laser Cut Architecture." *In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (***CHI'23***). *Hamburg, Germany. 2023.* [*Acceptance Rate: 28.39%*]

2022 [C.8] **Ruei-Che Chang**, Chao-Hsien Ting, Chia-Sheng Hung, Wan-Chen Lee, Liang-Jin Chen, Yu-Tzu Chao, Bing-Yu Chen, Anhong Guo. "OmniScribe: Authoring Immersive Audio Descriptions for 360° Videos." *In The 35th Annual ACM Symposium on User Interface Software and Technology* (***UIST'22***). *Bend, Oregon. 2022.* [*Acceptance Rate: 26.3%*]

[C.7] Ching-Wen Hung, **Ruei-Che Chang**, Hong-Sheng Chen, Chung-Han Liang, Liwei Chan, Bing-Yu Chen. "Puppeteer: Exploring Intuitive Hand Gestures and Upper-Body Postures for Manipulating Human Avatar Actions." *In The 28th Annual ACM Symposium on Virtual Reality Software and Technology* (***VRST'22***). *Tsukuba, Japan. 2022.* [*Acceptance Rate: 26.7%*]

2021 [C.6] **Ruei-Che Chang**, Chih-An Tsao, Fang-Ying Liao, Seraphina Yong, Tom Yeh, Bing-Yu Chen. "Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture." *In The 34th Annual ACM Symposium on User Interface Software and Technology* (***UIST'21***). *Virtual Event. 2021.* [*Acceptance Rate: 21%*]

[C.5] **Ruei-Che Chang***, Wen-Ping Wang*, Chi-Huan Chiang, Te-Yen Wu, Zheer Xu, Justin Luo, Bing-Yu Chen, Xing-Dong Yang. "AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision." *In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (***CHI'21***). *Virtual Event, Japan. 2021.* [*Acceptance Rate: 26.3%*]

2020 [C.4] **Ruei-Che Chang***, Chi-Huan Chiang*, Shuo-wen Hsu, Chih-Yun Yang, Da-Yuan Huang, Bing-Yu Chen. "TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays." *In Symposium on Spatial User Interaction* (***SUI'20***). *Virtual Event. 2020.* [*Acceptance Rate: 31%*]

[C.3] **Ruei-Che Chang***, Yi-Shyuan Chiang*, Yi-Lin Chuang, Shih-Ya Chou, Hao-Ping Lee, I-Ju Lin, Jian Hua Jiang Chen, Yung-Ju Chang. "Exploring the Design Space of User-System Communication for Smart-home Routine Assistants." *In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (***CHI'20***). *Virtual Event. 2020.* [*Acceptance Rate: 24.3%*]

[C.2] Kai-Chieh Huang, Chen-Kuo Sun, Da-Yuan Huang, Yu-Chun Chen, **Ruei-Che Chang**, Shuo-wen Hsu, Chih-Yun Yang, Bing-Yu Chen. "Glissade: Generating Balance Shifting Feedback to Facilitate Auxiliary Digital Pen Input." *In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (***CHI'20***). *Virtual Event. 2020.* [*Acceptance Rate: 24.3%*] **Best Paper Honorable Mention (Top 5% of 3126 submissions).**

2019 [C.1] Chi Wang, Da-Yuan Huang, Shuo-Wen Hsu, Chu-En Hou, Yeu-Luen Chiu, **Ruei-Che Chang**, Jo-Yu Lo, Bing-Yu Chen. "Masque: Exploring Lateral Skin Stretch Feedback on the Face with Head-Mounted Displays." *In Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (***UIST'19***). *New Orleans, LA. 2019.* [*Acceptance Rate: 24.4%*]

Peer-Reviewed Workshops, Posters, and Demos

2026 [A.7] Shuchang Xu*, Riku Arakawa*, Mina Huh*, Nandi Zhang*, Tianyu Zhang*, Wazeer Zulfikar*, **Ruei-Che Chang***, Yotam Sechayk*, Huamin Qu, Amy Pavel, Franklin Mingzhe Li, Yukang Yan, Brian A. Smith, Pattie Maes. "Cyber-Physical Systems for Accessibility and Ability Augmentation: Bridging Diverse Communities." *In Adjunct Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology* (***UIST'26***). *Detroit, MI. 2026.*

2025 [A.6] **Ruei-Che Chang**. "Enabling Real-World Assistive Agents: From Live Vision to Proactive Context-Aware Information Delivery." *In Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology* (***UIST'25 Doctoral Symposium***). *Busan, Korea. 2025.*

[A.5] **Ruei-Che Chang**. "Enabling Real-World Assistive Agents: From Live Vision to Proactive Context-Aware Information Delivery." *Accepted to 27th International ACM SIGACCESS Conference on Computers and Accessibility* (***ASSETS'25 Doctoral Consortium***). *Denver, Colorado. 2025.*

[A.4] Rosiana Natalie, Wenqian Xu, **Ruei-Che Chang**, Anhong Guo. "How Well Can Language Models Simulate the Vision Perception of People with Low Vision?" *In Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility* (***ASSETS'25 Poster***). *Denver, Colorado. 2025.*

2024 [A.3] **Ruei-Che Chang**, Yuxuan Liu, Anhong Guo. "Demonstration of WorldScribe: Towards Context-Aware Live Visual Descriptions." *In The 37th Annual ACM Symposium on User Interface Software and Technology* (***UIST'24***). *Pittsburgh, PA, USA. 2024.*

2023 [A.2] **Ruei-Che Chang**, Chia-Sheng Hong, Dhruv Jain, Anhong Guo. "SoundBlender: Exploring Sound Manipulations for Mixed-Reality Awareness." *In The 36th Annual ACM Symposium on User Interface Software and Technology* (***UIST'23 Demo***). *San Francisco, California. 2023.*

2022 [A.1] Ching-Wen Hung, **Ruei-Che Chang**, Hong-Sheng Chen, Chung-Han Liang, Liwei Chan, Bing-Yu Chen. "Puppeteer: Manipulating Human Avatar Actions with Intuitive Hand Gestures and Upper Body Postures." *In The 35th Annual ACM Symposium on User Interface Software and Technology* (***UIST'22 Poster***). *Bend, Oregon. 2022.*

Research Experiences Before PhD

Sep 2021 – Apr 2022 **Human-AI Lab, University of Michigan**
Research Intern, advised by Anhong Guo.
- Developed OmniScribe for authoring immersive audio descriptions for 360° videos [C.8].

Jun 2020 – Jul 2022 **Interactive Graphics Lab, National Taiwan University**
Visiting Student, advised by Bing-Yu Chen and Tom Yeh (University of Colorado).
- Developed Daedalus for non-visual accessible construction of laser-cut architecture [C.6].
- Conducted study to understand (non-)visual needs for laser-cut model design [C.9].

Feb 2019 – Aug 2019 **Interactive Graphics Lab, National Taiwan University**
Research Assistant, advised by Bing-Yu Chen.
- Developed Unity applications for Glissade [C.2] and Masque [C.1].
- Developed TanGo for expressive haptic interaction on VR headset [C.4].

Sep 2018 – Apr 2019 **Mobile and Ubiquitous Interaction Lab, National Yang Ming Chiao Tung University**
Research Assistant, advised by Yung-Ju (Stanley) Chang.
- Designed and conducted a study to understand the friction between humans and smart home [C.3].

Academic Services

Associate Chair, Program Committee
CHI LBW'23 '24 '25, UIST'26 Poster

Workshop Co-organizer
UIST'26 – Cyber-Physical Systems for Accessibility and Ability Augmentation

Organizing & Program Committee

Assistant to Program Chairs
UIST'26

Reviewing (100+ papers reviewed)

11 Special Recognitions for Outstanding Reviews: **CHI** '23 '24 '25 '26 | **UIST** '23 '24 '25 | **DIS** '24

Conferences
CHI('22 '23 '24 '25 '26), UIST('21 '22 '23 '24 '25 '26), DIS('22 '24), CSCW('23), TEI('23), SUI('23), ISS('22), MobileHCI('22), IEEE VR('23 '24), VRST('23), CHI LBW('20 '21 '22)

Journals
TACCESS('25), TOCHI('23)

Departmental & Other Service

Reviewer, PhD Admissions
Michigan AI Lab, 2025–2026

Student Volunteer
UIST'22

Invited Talk

Dec 2025 **Adobe HCI Seminar.** *"Supporting Context Switching When Multitasking with Computer Use Agents"*

Nov 2024 **CMU Accessibility Lunch Seminar.** *"Building a Real-World Assistant Agent for People who are blind"*

Nov 2024 **UMich CSE Honor Competition.** *"WorldScribe: Towards Context-Aware Live Visual Descriptions"*

Media Coverage

Nov 2024 **University of Michigan Engineering News.** "CSE Graduate Honors Competition showcases exceptional research by PhD students"

Oct 2024 **Health Tech World.** "AI tool gives blind person 'picture of the real world"'

Oct 2024 **University of Michigan Engineering News.** "Real-time descriptions of surroundings for people who are blind"

Teaching Experiences

Winter 2024 **EECS493 User Interface Development,** Graduate Student Instructor.

Mentorship

2025–2026 **Xirui Jiang**, *Undergrad student* at UMich. (*First Position*: MS student at UIUC)

2024–2026 **Wenqian Trista Xu**, *MS student* at UMich. (*First Position*: Data Analyst at GM Financial)

2024–2025 **Jovan Zheng Feng Yap**, *Undergrad* at UMich. (*First Position*: MS student at UC Berkeley)

2023–2024 **Yuxuan Liu**, *Undergrad* at UMich. (*First Position*: PhD student at UMich)

2023–2024 **Linfeng Song**, *Undergrad* at UMich. (*First Position*: MS student at UPenn)

2023–2024 **Andi Xu**, *Undergrad* at UMich. (*First Position*: MS Student at Stanford)

2022–2023 **Minyu Cai**, *Undergrad* at UMich. (*First Position*: MS student at CMU)

2022–2023 **Dier Hou**, *Undergrad* at UMich. (*First Position*: MS student at UCSD)

2022–2023 **Chia-Sheng Hung**, *MS student* at National Taiwan University. (*First Position*: Amazon, Taiwan)

2021–2022 **Fang-Ying Liao**, *MS student* at National Taiwan University. (*First Position*: Syntec, Taiwan)

2021–2022 **Chao-Hsien Ting**, *MS student* at National Taiwan University. (*First Position*: Realtek, Taiwan)

2021 **Chih-An Tsao**, *MS student* at National Taiwan University. (*First Position*: Sony, Taiwan)